2011-12-12

5011696401-PCA1

PCI-DMC-A01
PCI-DMC-B01

DMCNET™
Delta Motion Control Network

台達運動控制軸卡

(DMCNET™ 通訊型)

安裝說明

⚠ 注意事項

- ✔ 請在使用之前，詳細閱讀本安裝手冊。此安裝手冊只提供規格、伺服驅動器安裝及應用軟體(EzDMC)基本操作，有關於 PCI-DMC-A01 所包含應用與操作，在此無詳細的介紹，請讀者參閱相關的專業文章或書籍資料。
- ✔ 本通訊介面卡須搭配本公司 DMCNET 系列擴充模組產品或 ASDA-A2 系列伺服驅動器進行使用，實施配線時，務必拔除 / 切斷電源。
- ✔ 使用者使用本通訊介面卡時，必須將之安裝於具防塵、防潮及免於靜電破壞之工業電腦內。該電腦安裝的作業系統必須是 Windows 2000 / Windows XP。
- ✔ 輸入電源切斷後，一分鐘之內，請勿觸摸內部電路。請勿在上電時觸摸任何端子。
- ✔ 搭配使用的伺服驅動器上之接地端子 ⏚ 務必正確的接地，可提高產品抗雜訊能力。
- ✔ 通訊介面卡上連接口之端子禁止外力重壓，避免導致產品損壞。

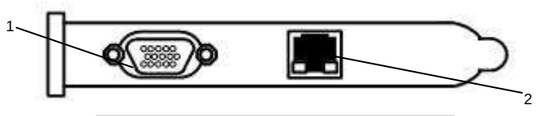
❶ 產品簡介

1.1 型號說明

PCI	-	DMC	-	A01
(1)		(2)		(3)

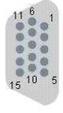
(1) 產品介面	PCI 介面
(2) 產品名稱	DMC：Delta Motion Control
(3) 版本識別碼	A01 ver.1.1：標準品 (DI x1, DO x1) A01 ver.2.0：標準品 (DI x8, DO x4) B01 ：標準品 (DI x1, DO x1, 高速計數器 x2)

1.2 連接埠介紹



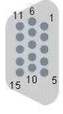
1. 輸入 / 輸出訊號連接埠 (DSUB15)
2. DMCNET 擴充模組連接埠

PCI-DMC-A01 Ver. 2.0
DSUB15 腳位定義



Pin	標記	Pin	標記
1	GPIO IN 0	9	GPIO IN 6
2	GPIO IN 1	10	GPIO IN 7
3	GPIO IN 2	11	External GND
4	GPIO IN 3	12	GPIO OUT 0
5	External GND	13	GPIO OUT 1
6	E24V	14	GPIO OUT 2
7	GPIO IN 4	15	GPIO OUT 3
8	GPIO IN 5		

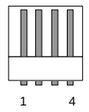
PCI-DMC-B01 Ver. 2.0
PCI-DMC-A01 Ver. 1.1
DSUB15 腳位定義



Pin	標記	Pin	標記
1	QEP_A_1-	9	QEP_B_2+
2	QEP_B_1-	10	GPIO IN
3	QEP_A_2-	11	QEP_CMP_1+(RS-422)
4	QEP_B_2-	12	QEP_CMP_1- (RS-422)
5	External GND	13	QEP_CMP_2+(RS-422)
6	QEP_A_1+	14	QEP_CMP_2- (RS-422)
7	QEP_B_1+	15	GPIO OUT
8	QEP_A_2+		

※ PCI-DMC-A01 Ver. 1.1 僅 **Pin 5**、**Pin 10**、**Pin 15** 腳位為**有效**。

PCI-DMC-B01
Compare Output
腳位定義



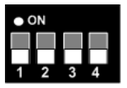
Pin	標記	說明
1	CMP_OUT1(QEP1)	CMOS 3.3V 到位比較觸發訊號輸出
2	GND	GND 訊號
3	CMP_1+(LVDS)	LVDS 差動訊號 (+)
4	CMP_1- (LVDS)	LVDS 差動訊號 (-)

CN9 與 CN10 為 PCI-DMC-B01 專用。PCI-DMC-A01 則無此料件。

※此為座端的腳位定義。線端的 PIN 1 會對應此處的 PIN 4。

1.3 PCI-DMC-A01 版本辨視

Ver. 1.1



Pin	標記	說明
1	Bit3	Card ID Bit3
2	Bit2	Card ID Bit2
3	Bit1	Card ID Bit1
4	Bit0	Card ID Bit0

Card ID= Bit3x2*3+Bit2x2*2+Bit1x2*1+Bit0x2*0

Ver. 2.0



Card ID 為旋鈕調撥位置對應的數值。
其設定的數值範圍為 0~15。

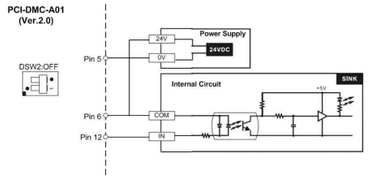
❷ 功能規格

選　　項	PCI-DMC-A01 v1.1	PCI-DMC-A01 v2.0	PCI-DMC-B01 v2.0
支援驅動器模組	Delta ASDA-A2-F 伺服驅動器		
原點復歸模式	35 種 (藉由 DMCNET 參數設置控制)		
運動速率切線控制	T-curve、S-curve		
補間模式	線性、圓弧與連續		
Ring 數	1 組		
傳輸線	Category 5e STP Ethernet cable (24AWG/4Pairs)		
通訊距離	最大 30 公尺(12 slave module)		
串列控制介面	half duplex RS-485 with transformer isolation		
PCI 規格	ver.2.2；支援 32-bit, 3.3V/5VDC 操作型式		
電源消耗量	+5V DC at 0.5A typical		
雜訊免疫力	突波電壓耐受量：1500VAC(Primary-secondary)；1500VAC(Primary-PE) ESD (IEC 61131-2, IEC 61000-4-2): 8KV Air Discharge EFT (IEC 61131-2, IEC 61000-4-4): Power Line: 2KV, Communication I/O: 1KV RS (IEC 61131-2, IEC 61000-4-3): 26MHz ~ 1GHz, 10V/m		
操作環境	操作溫度：0℃ ~ 50℃；儲存溫度：-20℃ ~ 70℃		
	濕度：5 ~ 95% RH 以下 (不結露)		
DI 數	1	8	1
DO 數	1	4	1
高速計數器	-	-	2

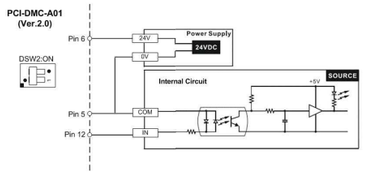
❸ 接線範例

❖ PCI-DMC-A01 連結 Input 裝置

SINK

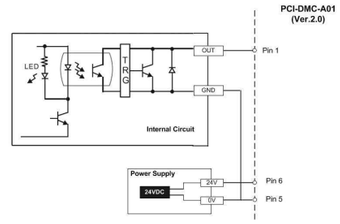


SOURCE



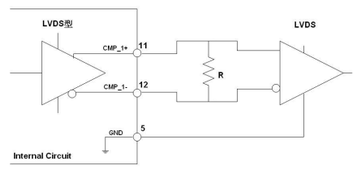
❖ 外部 Output 裝置連結 PCI-DMC-A01

SINK

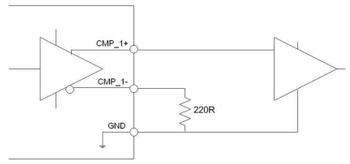


❖ PCI-DMC-B01 高速計數器之接線

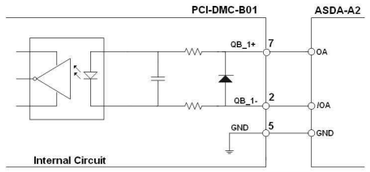
Compare(RS-422 差動式)



Compare(非差動式)

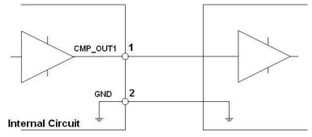


Encoder



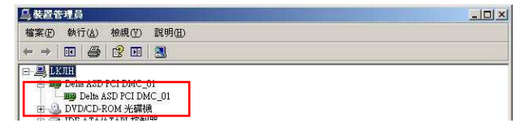
❖ PCI-DMC-B01 3.3V Compare Output

CN9 & CN10



❹ 驅動程式安裝與應用程式使用

❖ 安裝驅動與應用程式軟體，安裝完畢後，檢視「裝置管理員」中「Delta ASD PCI DMC_01」是否有被新增。



❖ 應用程式 EzDMC 的開啟與使用

點選「開始」→「程式集」→「Delta Industrial Automation」→「PCI-DMC-A01」，點選執行【EzDMC】。



NOTE

1) 本說明書內所記載之規格若有變更，本公司恕不另行通知。當內容規格有所修正時，請洽詢代理商或至台達網站 http://www.delta.com.tw/industrialautomation/下載最新版本。

2011-12-12

5011696401-PCA1

PCI-DMC-A01
PCI-DMC-B01

DMCNET™
Delta Motion Control Network

台达运动控制轴卡

(DMCNET™ 通讯型)

安装说明

⚠ 注意事项

- ✓ 请在使用之前，详细阅读本安装手册。此安装手册只提供规格、伺服驱动器安装及应用软件(EzDMC)基本操作，有关于 PCI-DMC-A01 所包含应用与操作，在此无详细的介绍，请读者参阅相关的专业文章或书籍资料。
- ✓ 本通讯界面卡须搭配本公司 DMCNET 系列扩展模块产品或 ASDA-A2 系列伺服驱动器进行使用，实施配线时，务必拔除 / 切断电源。
- ✓ 使用者使用本通讯界面卡时，必须将其安装于具防尘、防潮及免于静电破坏的工业电脑内，该电脑安装的作业系统必须是 Windows 2000 / Windows XP。
- ✓ 输入电源切断后，一分钟之内，请勿触摸内部电路。请勿在上电时触摸任何端子。
- ✓ 搭配使用的伺服驱动器上的接地端子 ⏚ 务必正确的接地，可提高产品抗噪音能力。
- ✓ 通讯界面卡上连接口的端子禁止外力重压，避免导致产品损坏。

❶ 产品简介

1.1 型号说明

PCI (1) - DMC (2) - A01 (3)

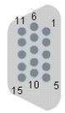
(1) 产品接口	PCI 接口
(2) 产品名称	DMC：Delta Motion Control
(3) 版本识别码	A01 ver.1.1：标准品 (DI x1, DO x1) A01 ver.2.0：标准品 (DI x8, DO x4) B01 ：标准品 (DI x1, DO x1, 高速计数器 x2)

1.2 连接口说明



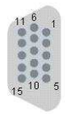
1. 输入 / 输出信号连接端口
2. DMCNET 扩展模块连接端口

PCI-DMC-A01 Ver. 2.0
DSUB15 脚位定义



Pin	标记	Pin	标记
1	GPIO IN 0	9	GPIO IN 6
2	GPIO IN 1	10	GPIO IN 7
3	GPIO IN 2	11	External GND
4	GPIO IN 3	12	GPIO OUT 0
5	External GND	13	GPIO OUT 1
6	E24V	14	GPIO OUT 2
7	GPIO IN 4	15	GPIO OUT 3
8	GPIO IN 5		

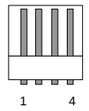
PCI-DMC-B01 Ver. 2.0
PCI-DMC-A01 Ver. 1.1
DSUB15 脚位定义



Pin	标记	Pin	标记
1	QEP_A_1-	9	QEP_B_2+
2	QEP_B_1-	10	GPIO IN
3	QEP_A_2-	11	QEP_CMP_1+(RS-422)
4	QEP_B_2-	12	QEP_CMP_1- (RS-422)
5	External GND	13	QEP_CMP_2+(RS-422)
6	QEP_A_1+	14	QEP_CMP_2- (RS-422)
7	QEP_B_1+	15	GPIO OUT
8	QEP_A_2+		

※PCI-DMC-A01 Ver. 1.1 仅 **Pin 5**、**Pin 10**、**Pin 15** 脚位为**有效**。

PCI-DMC-B01
Compare Output
脚位定义



Pin	标记	说明
1	CMP_OUT1(QEP1)	CMOS 3.3V 到位比较触发信号输出
2	GND	GND 信号
3	CMP_1+(LVDS)	LVDS 差动信号 (+)
4	CMP_1- (LVDS)	LVDS 差动信号 (-)

CN9 与 CN10 为 PCI-DMC-B01 专用。PCI-DMC-A01 则无此料件。
※此为座端的脚位定义。线端的 PIN 1 会对应此处的 PIN 4。

1.3 PCI-DMC-A01 版本辨视

Ver. 1.1



Card ID= Bit3x2*3+Bit2x2*2+Bit1x2*1+Bit0x2*0

Pin	标记	说明
1	Bit3	Card ID Bit3
2	Bit2	Card ID Bit2
3	Bit1	Card ID Bit1
4	Bit0	Card ID Bit0

Ver. 2.0



Card ID 为旋钮调拨位置对应的数值。
其设定的数值范围为 0~15。

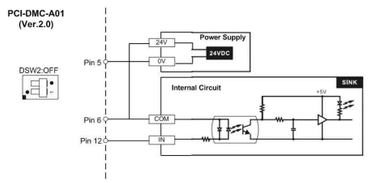
❷ 功能规格

选 项	PCI-DMC-A01 v1.1	PCI-DMC-A01 v2.0	PCI-DMC-B01 v2.0
支持驱动器模块	Delta ASDA-A2-F 伺服驱动器		
原点复位模式	35 种 (借由 DMCNET 参数设置控制)		
运动速率切线控制	T-curve、S-curve		
插补模式	线性、圆弧与连续		
Ring 数	1 组		
传输线	Category 5e STP Ethernet cable (24AWG/4Pairs)		
通讯距离	最大 30 公尺(12 slave module)		
串行控制接口	half duplex RS-485 with transformer isolation		
PCI 规格	ver.2.2；支持 32-bit, 3.3V/5VDC 操作型式		
电源消耗量	+5V DC at 0.5A typical		
抗噪音能力	突波电压耐受量：1500VAC(Primary-secondary)；1500VAC(Primary-PE) ESD (IEC 61131-2, IEC 61000-4-2): 8KV Air Discharge EFT (IEC 61131-2, IEC 61000-4-4): Power Line: 2KV, Communication I/O: 1KV RS (IEC 61131-2, IEC 61000-4-3): 26MHz ~ 1GHz, 10V/m		
操作环境	操作温度：0℃ ~ 50℃；储存温度：-20℃ ~ 70℃ 湿度：5 ~ 95% RH 以下 (不结露)		
DI 数	1	8	1
DO 数	1	4	1
高速计数器	-	-	2

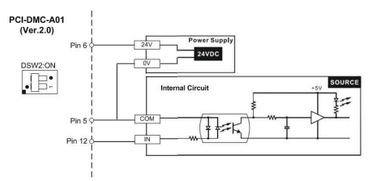
❸ 安装及配线

❖ PCI-DMC-A01 连结 Input 装置

SINK

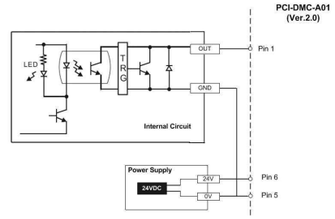


SOURCE



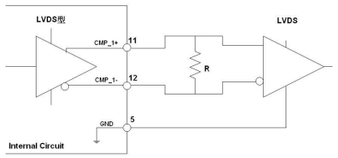
❖ 外部 Output 装置连结 PCI-DMC-A01

SINK

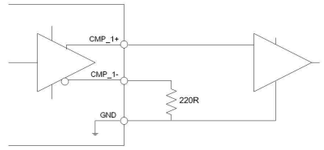


❖ PCI-DMC-B01 高速计数器的接线

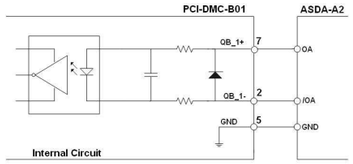
Compare(RS-422 差动式)



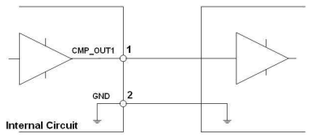
Compare(非差动式)



Encoder

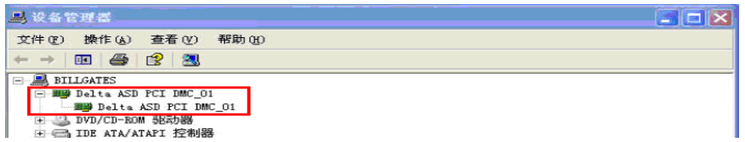


❖ PCI-DMC-B01 3.3V Compare Output
CN9 & CN10



❹ 驱动程序安装与应用程序使用

❖ 安装驱动与应用程序软件，安装完毕后，检查「设备管理器」中「Delta ASD PCI DMC_01」是否有被新增。



❖ 应用程序 EzDMC 的开启与使用

点击「开始」→「程序」→「Delta Industrial Automation」→「PCI-DMC-A01」，点击执行【EzDMC】。



NOTE

本说明书内所记载的规格若有变更，本公司恕不另行通知，当内容规格有所修正时，请咨询代理商或至台达网站 http://www.delta.com.tw/industrialautomation/ 下载最新版本。

2011-12-12

5011696401-PCA1

PCI-DMC-A01
PCI-DMC-B01

Delta Motion Control Card (DMCNET™ Communication Type)

DMCNET™ Delta Motion Control Network

Instruction Sheet

Warning

- ✓ Please read this instruction carefully before use.
- ✓ This instruction provides the information of specifications, installation and basic operation and settings of EzDMC application software program. Regarding the applications of PCI-DMC-A01 and PCI-DMC-B01 DMCNET motion control cards, there is no detailed information available on this instruction. If the users need complete information of the applications of PCI-DMC-A01 and PCI-DMC-B01 DMCNET motion control cards, please refer to relevant professional articles and books.
- ✓ PCI-DMC-A01 and PCI-DMC-B01 DMCNET motion control cards should be used with DMCNET extension modules or ASDA-A2 series servo system. Ensure to switch off the power before wiring.
- ✓ Please install this PCI-DMC-A01 and PCI-DMC-B01 DMCNET motion control cards in an industrial embedded PC free of airborne dust, humidity and static electricity. The operating system must be Windows 2000 or Windows XP.
- ✓ DO NOT touch any internal circuit in 1 minute after the power is switched off. Do NOT touch any terminal when the power is switched on.
- ✓ Make sure the ground terminal ⏚ is correctly grounded in order to prevent electromagnetic interference.
- ✓ DO NOT place any heavy objects on the connection port of PCI-DMC-A01 and PCI-DMC-B01 DMCNET motion control cards. Doing so may damage the product.

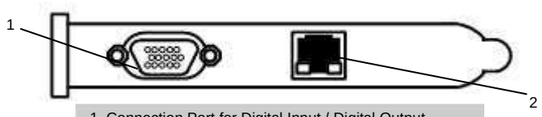
❶ Introduction

1.1 Model Explanation

PCI	-	DMC	-	A01
(1)		(2)		(3)

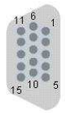
(1) Product Interface	PCI Interface
(2) Product Name	DMC: Delta Motion Control
(3) Version	A01 ver.1.1 : Standard (1 digital input, 1 digital output) A01 ver.2.0 : Standard (8 digital inputs, 4 digital outputs) B01 : Standard (1 digital input, 1 digital output, 2 high-speed counters)

1.2 Connection Port Selection



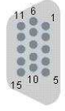
1. Connection Port for Digital Input / Digital Output
2. Connection Port for DMCNET Extension Module

PCI-DMC-A01 Ver. 2.0
DSUB15 PIN
Definition



Pin No.	Identification	Pin No.	Definition
1	GPIO IN 0	9	GPIO IN 6
2	GPIO IN 1	10	GPIO IN 7
3	GPIO IN 2	11	External GND
4	GPIO IN 3	12	GPIO OUT 0
5	External GND	13	GPIO OUT 1
6	E24V	14	GPIO OUT 2
7	GPIO IN 4	15	GPIO OUT 3
8	GPIO IN 5		

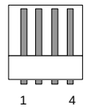
PCI-DMC-B01 Ver. 2.0
PCI-DMC-A01 Ver. 1.1
DSUB15 PIN
Definition



Pin No.	Definition	Pin No.	Definition
1	QEP_A_1-	9	QEP_B_2+
2	QEP_B_1-	10	GPIO IN
3	QEP_A_2-	11	QEP_CMP_1+(RS-422)
4	QEP_B_2-	12	QEP_CMP_1- (RS-422)
5	External GND	13	QEP_CMP_2+(RS-422)
6	QEP_A_1+	14	QEP_CMP_2- (RS-422)
7	QEP_B_1+	15	GPIO OUT
8	QEP_A_2+		

※In PCI-DMC-A01 Ver. 1.1, only **Pin 5, Pin 10 and Pin 15 are available for use.**

PCI-DMC-B01
Compare Output
Pin Definition



Pin No.	Definition	Description
1	CMP_OUT1(QEP1)	CMOS 3.3V Compare Output
2	GND	GND Signal
3	CMP_1+(LVDS)	LVDS Differential Signal (+)
4	CMP_1- (LVDS)	LVDS Differential Signal (-)

CN9 and CN10 connectors are used for PCI-DMC-B01model only, not for PCI-DMC-A01model.

※This is the pin definition of socket. The pin 1 of cable will be corresponding to the pin 4 of this socket.

1.3 Swiches of PCI-DMC-A01

Ver. 1.1



Pin No.	Definition	Description
1	Bit3	Card ID Bit3
2	Bit2	Card ID Bit2
3	Bit1	Card ID Bit1
4	Bit0	Card ID Bit0

Card ID= Bit3x2^{*3}+Bit2x2^{*2}+Bit1x2^{*1}+Bit0x2^{*0}

Ver. 2.0



Card ID indicates the value of the corresponding switch position. Its setting range is within 0~15.

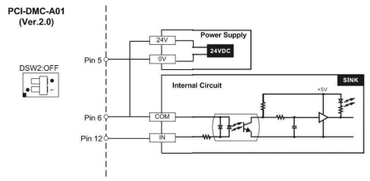
❷ Specifications

Item	PCI-DMC-A01 v1.1	PCI-DMC-A01 v2.0	PCI-DMC-B01 v2.0
Supporting Module	Delta ASDA-A2-F Type Servo Drive		
Homing Mode	35 types (Parameter Setting via DMCNET)		
Velocity Profiles	T-curve, S-curve		
Interpolation Mode	Linear, Circular and Continuous		
Ring	1 Ring		
Communication Cable	Category 5e STP Ethernet cable (24AWG/4Pairs)		
Communication Distance	Max. 30m (12 slave module)		
Communication Interface	half duplex RS-485 with transformer isolation		
PCI Specifications	ver.2.2, supports 32-bit, 3.3V/5VDC operation		
Power Consumption	+5V DC at 0.5A typical		
Noise Tolerance Threshold	Withstand (Peak) voltage: 1500VAC(Primary-secondary); 1500VAC(Primary-PE) ESD (IEC 61131-2, IEC 61000-4-2): 8KV Air Discharge EFT (IEC 61131-2, IEC 61000-4-4): Power Line: 2KV, Communication I/O: 1KV RS (IEC 61131-2, IEC 61000-4-3): 26MHz ~ 1GHz, 10V/m		
Environment	Operating Temperature: 0℃ ~ 50℃; Storage Temperature: -20℃ ~ 70℃		
	Humidity: 5 ~ 95% (non-condensing)		
Number of Digital Input	1	8	1
Number of Digital Output	1	4	1
Number of High-Speed Counters	-	-	2

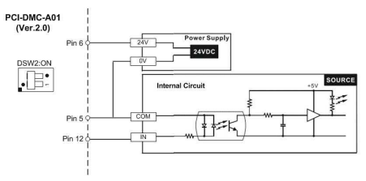
❸ Wiring

❖ Input Circuit Wiring of PCI-DMC-A01

SINK

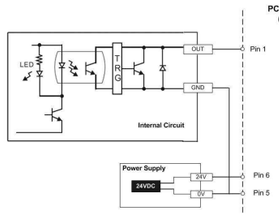


SOURCE



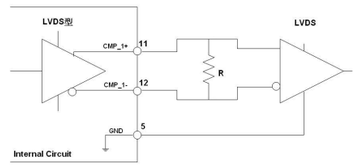
❖ Output Circuit Wiring of PCI-DMC-A01

SINK

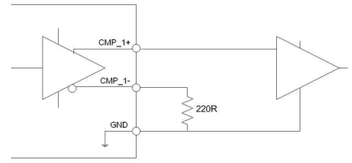


❖ PCI-DMC-B01 Wiring of High-Speed Counters

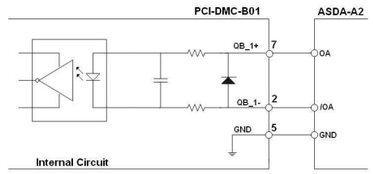
Compare(RS-422 Differential)



Compare(In-differential)

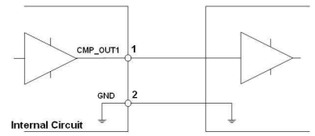


Encoder



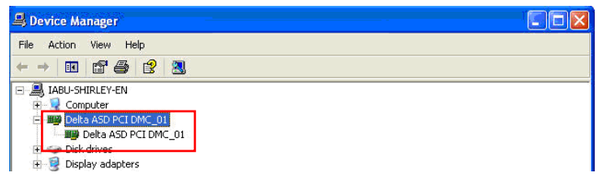
❖ PCI-DMC-B01 3.3V Compare Output

CN9 & CN10



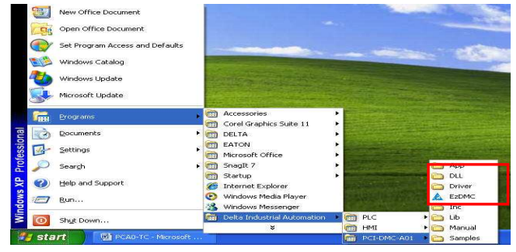
❹ Software Settings

❖ Install PCI-DMC-A01 setup driver software first. After setup, open “Device Manager” dialog box to check if PCI-DMC-A01 DMCNET motion control card (Delta ASD PCI DMC_01) is connected or not.



❖ Activate and Use EzDMC Application Software Program

Click “start” → “Programs” → ”Delta Industrial Automation” → “PCI-DMC-A01”, and then execute [EzDMC] to activate EzDMC application software program.



NOTE

1) The content of this instruction sheet may be revised without prior notice. Please consult our distributors or download the most updated version at http://www.delta.com.tw/industrialautomation.